



Week 5 Relational Model

ISB02503202 - Database

Even Semester 2025/2026
3 SKS

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Relational Model

- Introduced in 1970
- By Edgar F. Codd
 - was engineer in IBM
 - This model using a mathematical expression known as "Relational Algebra"
- Currently is a standard model for the DBMS products

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Terms in Relational Model

- Table (Relation)
- Functional Dependency
- Determinant
- Candidate Key
- Composite Key
- Primary Key
- Surrogate Key
- Foreign Key
- Referential Integrity Constraint
- Normal Form
- Multivalued Dependency

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Relation (= Table*)

- Is set of some **tuples** (data)
- A model has **column** (kolom) and **row** (baris)

EmployeeNumber	FirstName	LastName	Department	Email	Phone
100	Jerry	Johnson	Accounting	JJ@somewhere.com	834-1101
200	Mary	Abemathy	Finance	MA@somewhere.com	834-2101
300	Liz	Smathers	Finance	LS@somewhere.com	834-2102
400	Tom	Caruthers	Accounting	TC@somewhere.com	834-1102
500	Tom	Jackson	Production	TJ@somewhere.com	834-4101
600	Eleanore	Caldera	Legal	EC@somewhere.com	834-3101
700	Richard	Bandalone	Legal	RB@somewhere.com	834-3102

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Characteristics Relation (Table)

- A row contained 1 record/data stored in a entity (*entity instances*)
- A column contained **attribute** of entity
- All data on the same column **has same data type**
- Each column has **unique name** (can't be the same)
- 1 cell on the table contained **1 single value** (can't be multivalue)
- The order in the column and row doesn't matters
- No 2 rows can be the same

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Example NOT Relation (table) WHY??

EmployeeNumber	FirstName	LastName	Department	Email	Phone
100	Jerry	Johnson	Accounting	JJ@somewhere.com	834-1101
200	Mary	Abemathy	Finance	MA@somewhere.com	834-2101
300	Liz	Smathers	Finance	LS@somewhere.com	834-2102
400	Tom	Caruthers	Accounting	TC@somewhere.com	834-1102, 834-1191, 834-1192
500	Tom	Jackson	Production	TJ@somewhere.com	834-4101
600	Eleanore	Caldera	Legal	EC@somewhere.com	834-3101
700	Richard	Bandalone	Legal	RB@somewhere.com	834-3102, 834-3191

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Another Terms

Table	Column	Row
Relation	Attribute	Tuple
File	Field	Record

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Functional Dependency

- *Functional dependency* appears when the value of one attribute determines value of another attribute.
- E.g.:
 - ObjectColor □ Weight
 - ObjectColor □ Shape
- Read: ObjectColor determine Weight and Shape

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Determinant / Dependent

- Attribute the left side on the functional dependency called **determinant**
- Attribute the right side on the functional dependency called **dependent**

ObjectColor □ **(Weight, Shape)**
(determinant) (dependent)

How to Determine Functional Dependency (FD)

	SKU	SKU_Description	Department	Buyer
1	100100	Std. Scuba Tank, Yellow	Water Sports	Pete Hansen
2	100200	Std. Scuba Tank, Magenta	Water Sports	Pete Hansen
3	100300	Std. Scuba Tank, Light Blue	Water Sports	Pete Hansen
4	100400	Std. Scuba Tank, Dark Blue	Water Sports	Pete Hansen
5	100500	Std. Scuba Tank, Light Green	Water Sports	Pete Hansen
6	100600	Std. Scuba Tank, Dark Green	Water Sports	Pete Hansen
7	101100	Dive Mask, Small Clear	Water Sports	Nancy Meyers
8	101200	Dive Mask, Med Clear	Water Sports	Nancy Meyers
9	201000	Half-dome Tent	Camping	Cindy Lo
10	202000	Half-dome Tent Vestibule	Camping	Cindy Lo
11	203000	Half-dome Tent Vestibule - Wide	Camping	Cindy Lo
12	301000	Light Fly Climbing Harness	Climbing	Jerry Martin
13	302000	Locking Carabiner, Oval	Climbing	Jerry Martin

Answer FD (1)

- SKU □ (SKU_Description, Department, Buyer)
- SKU_Description □ (SKU, Department, Buyer)
- Buyer □ Department

Another Example

	OrderNumber	SKU	Quantity	Price	ExtendedPrice
1	1000	201000	1	300.00	300.00
2	1000	202000	1	130.00	130.00
3	2000	101100	4	50.00	200.00
4	2000	101200	2	50.00	100.00
5	3000	100200	1	300.00	300.00
6	3000	101100	2	50.00	100.00
7	3000	101200	1	50.00	50.00

Answer FD (2)

- (OrderNumber, SKU) □ (Quantity, Price, ExtendedPrice)
- (Quantity, Price) □ ExtendedPrice

5 different types of Key

- Candidate Key □ Key can determine another column in a relation
- Composite Key □ Key consisting of 2 or more column
- Primary Key □ Candidate Key selected as the primary key to identify row
- Surrogate Key □ the additional Column on the relation acts as primary key
- Foreign Key □ Key is connecting between relation

Key is combination of one or more column used to **identify** a row in a relation

Primary Key

- Candidate Key is selected as the primary to identify the row
- Rules :
 - Only exist 1 primary key in a relation
 - Primary key could be composite key
 - Primary key the ideal is the value is short, numeric, and never changes

Surrogate Key

- A additional Column on the that acts as the primary key
- Rules :
 - The value is short, numeric, and never (same as primary key)
 - Has meaningless value to user
 - Usually not shown on the report and form

Example of Surrogate Key

RENTAL_PROPERTY *without* surrogate key:

RENTAL_PROPERTY (Street, City, State/Province, Zip/PostalCode,
Country, Rental_Rate)

RENTAL_PROPERTY **with** surrogate key:

RENTAL_PROPERTY (PropertyID, Street, City, State/Province,
Zip/PostalCode, Country, Rental_Rate)

bold,underline = primary key
italic = foreign key

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Foreign Key

- Primary Key on the relation and resides in another relation with the purpose to connecting between two relation
- Foreign key can single column or is composite key
- The Term foreign key refers to the value of a key foreign to the relation where the key appears as a foreign key value

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Example of Foreign Key

DEPARTMENT:

DEPARTMENT (DepartmentName, BudgetCode, ManagerName)

EMPLOYEE dengan foreign key:

EMPLOYEE (EmployeeNumber, EmployeeName, *DepartmentName*)

bold,underline = primary key
italic = foreign key

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Referential Integrity Constraint

Referential Integrity Constraint is term that limits the value of foreign key that must exist as the primary key on the original relation

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Normal Form

- Relations are not the same as each other.
- Relations are categorized into normal forms based on the types of problems they address.
- Categories of normal forms:
 - 1NF
 - 2NF
 - 3NF
 - BCNF
 - 5NF
 - DK/NF

Further discussion in
Week 06.

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Summary of Normalization Theory

Source of Anomaly	Normal Forms	Design Principles
Functional dependencies	1NF, 2NF, 3NF, BCNF	BCNF: Design tables so that every determinant is a candidate key.
Multivalued dependencies	4NF	4NF: Move each multivalued dependency to a table of its own.
Data constraints and oddities	5NF, DK/NF	DK/NF: Make every constraint a logical consequence of candidate keys and domains.

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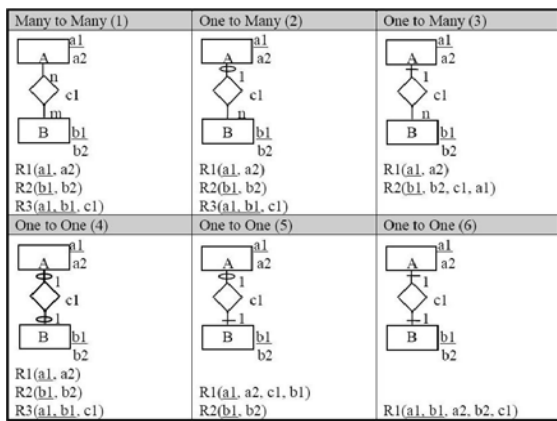
Convert ER Model To Relational Model

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Intro

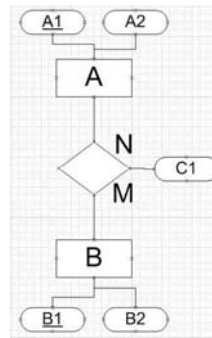
- Process convert from the E.R. model to Relational Model can be done by following the six rules seen from its cardinality
- The goal is to avoid non-homogeneous relation.
- Process conversion is done sequentially from rules 1 to 6

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Rule #1 for Many-to-Many (N-M)

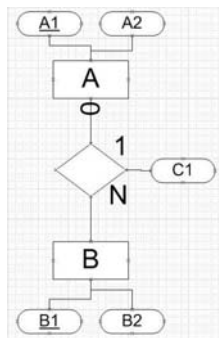


- A (A1, A2)
- B (B1, B2)
- A_B (A1B1, C1)

bold.underline = primary key
italic = foreign key

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Rule #2 for One-to-Many (1-N) [Min. Cardinality 0 on the One side]

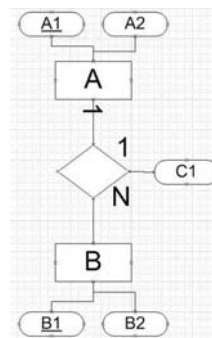


- A (A1, A2)
- B (B1, B2)
- A_B (A1B1, C1)

bold.underline = primary key
italic = foreign key

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Rule #3 for One-to-Many (1-N) [Min. Cardinality 1 on the One side]

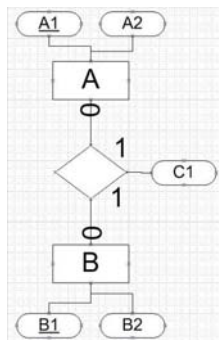


- A (A1, A2)
- B (B1, B2, C1, A1)

bold.underline = primary key
italic = foreign key

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Rule #4 for One-to-One (1-1) [Min. Cardinality 0 on both sides]

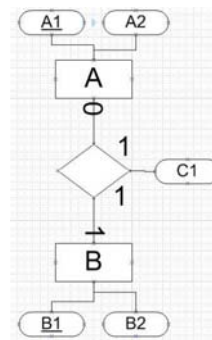


- A (A1, A2)
- B (B1, B2)
- A_B (A1B1, C1)

bold.underline = primary key
italic = foreign key

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Rule #5 for One-to-One (1-1) [Min. Cardinality 0 and 1]

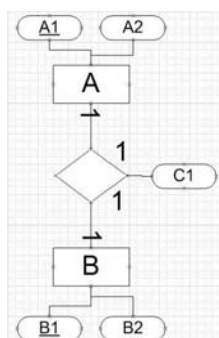


- A (A1, A2, C1, B1)
- B (B1, B2)

bold.underline = primary key
italic = foreign key

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Rule #6 for One-to-One (1-1) [Min. Cardinality 1 on both sides]

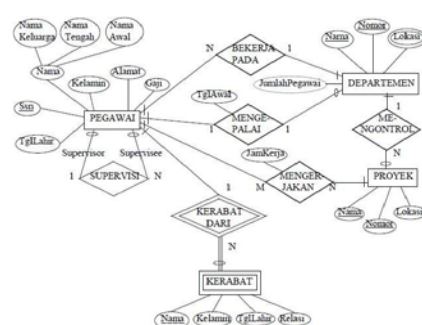


- A_B (A1B1, A2, B2, C1)

bold.underline = primary key
italic = foreign key

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Example (ER Model)



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Result (the Relational Model)

- PEGAWAI (**SsnPeg**, Nama, Alamat, JenisKelamin, TglLahir, Gaji, *NomorDep*)
- SUPERVISI (**SsnSupervisor**, *SsnSupervisee*)
- DEPARTEMEN (NamaDep, **NomorDep**, Lokasi, Jumlah, *SsnKepala*, TglAwal)
- PROYEK (NamaProyek, **NomorProyek**, Lokasi, *NomorDep*)
- MENGERJAKAN (**NamaProyek**, **SsnPeg**)
- KERABAT (**NomorKerabat**, NamaKerabat, JenisKelamin, TglLahir, Relasi, *SsnPeg*) **NomorKerabat is a surrogate key*

bold,underline = primary key
italic = foreign key



THANK YOU

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